



Microwave irradiation-based rapid curing of PDMS for microfluidic applications

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Abstract

Microfluidic devices are routinely fabricated using replica moulding. The master relief structure is generally constructed on a silicon substrate and the stamps are made using polydimethylsiloxane (PDMS), cured via the slow process of hot-air convective heating. An alternative method that greatly reduces this cure time is presented here. It employs a low-cost, consumer-grade microwave oven and speeds up the curing process by permitting a much higher heat flux to be applied to the prepolymer. A specific pulsed irradiation protocol has been devised which achieves precisely controlled energy delivery for this purpose. This is necessary because silicon's relatively much stronger microwave absorption causes uncontrollable temperature overshoots that could degrade PDMS if continuously irradiated. The protocol has been arrived at after due optimization of operational parameters, achieved via simulation of a finite element model. It has been implemented by modifying the control circuitry of the consumer grade oven. Cure times as low as two minutes have been demonstrated. Optical and mechanical characterization results show no adverse effect on the performance of PDMS cured using the proposed method. Thus, it can be used for the rapid fabrication of PDMS-based microfluidic devices and would be especially beneficial for research labs looking to speed up microfluidic device development iterations.

Keywords Rapid fabrication · PDMS · PDMS curing · Soft lithography

1 Introduction

The field of microfluidics has demonstrated tremendous potential over the last 3 decades. The elastomer polydimethylsiloxane (PDMS) has been the focal material for prototyping as well as production of microfluidic devices (Whitesides 2006). The bio-compatibility, ease of handling and low cost of this thermosetting polymer make it the material of choice for soft-lithography. However, the long cure time (running into several hours) discourages rapid prototyping using PDMS (Hum 2006). This is a drawback that especially affects the research community since it slows down development iterations. Moreover, there is still a significant equipment cost involved due to the requirement of a hot-air oven for thermal curing. This cost becomes important when laboratories in developing countries need to work in this domain. It is so because microfluidics finds its major applications in

resource-challenged settings such as bio-medical diagnostics in developing countries. Hence, the researchers in these places would benefit from a method of PDMS curing that is quick and incurs as little expense as possible.

PDMS curing or polymerization (cross-linking process) is catalyzed by either a peroxide-based catalyst or a platinum based one (Wacker 2014). In both cases, the polymerization requires substantial activation energy which is provided by a thermal energy source. Conventionally, a hot-air convection system is used. An electric element heats the air which then provides the heat flux to the PDMS. A minimum of 2 h or more is needed for the curing, though the protocols followed in labs often call for an overnight cure at 60 or 65 °C. A much higher heat flux can be generated via microwave absorption, as per the effect of dielectric heating. Heat is produced by rapid oscillations of dipoles of the insulating material upon interaction with an oscillating RF field. If the frequency of field is high, the dipoles cannot align quickly enough and the phase lag results in dissipation of energy as heat. The thermal power absorbed per unit volume (Linn and Möller 2005) is dependent on the strength of the electromagnetic wave E (dependent on microwave power), its

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exciting frequency f and the imaginary part of the material's relative dielectric constant ϵ'' , as can be seen in the following equation:

$$P = 2\pi f \epsilon_0 \epsilon'' E^2, \quad (1)$$

where ϵ_0 is permittivity of free space.

Table 1 compares the values of the two components of dielectric constant of three materials (relevant to this work): silicon as the substrate of the master, glass as the material of the Petri dish used to hold the silicon wafer and PDMS as the microfluidic substrate.

It can be noticed that ϵ''_{Si} is far higher than both, ϵ''_{PDMS} and ϵ''_{Glass} . Thus, if a silicon-based master containing PDMS is irradiated, the silicon layer will absorb almost all the microwave power. It will act as the heat source and conduct thermal energy to the PDMS layer in contact. This conductive heating still follows the Fourier's law depicted in Eq. 1. However, the thermal conductivity k of silicon [48 W/K m (Slack and Glassbrenner 1964)] is much higher than that of air [about 0.1 W/K m (Lemmon and Jacobsen 2004)]. Also, this heat transfer is by conduction, i.e., the heat source is in contact and hence the thermal gradient $\frac{dT}{dx}$ is much higher. Thus, the enhanced heat flux $\dot{q}$ leads to faster curing.

While the utility of microwaves in polymer science is known for a while (Wiesbrock et al. 2004; Sinnwell and Ritter 2007), their application has been restricted to industrial and research purposes such as development of novel polymer coatings (Choi et al. 2017) and polymer composites (Uyanik et al. 2006). Though Gojo et al. (2008) and Morbioli et al. (2018) touched upon the aspect of microwaves for curing PDMS on a glass substrate, neither the theory nor any experimental protocol has been reported for use in replica moulding for microfluidic devices with a silicon substrate. Also, in their case, the glass substrate allowed straightforward usage of a microwave oven, because ϵ''_{Glass} compares well with ϵ''_{PDMS} and thus a bulk of the microwave energy is absorbed directly by the PDMS. In contrast, in a silicon-PDMS combine, it is the silicon that absorbs most of the energy. Thus, the microwave irradiation needs to be precisely controlled to avoid temperature overshoots in the silicon layer which could otherwise degrade the PDMS and hence the master relief structures. A simple technique to

achieve this is reported here, as described in the following text. Thus, to the best of our knowledge, this is the first reported protocol of microwave-based curing of PDMS for replica moulding to fabricate microfluidic devices.

1.1 Pulsed microwave irradiation

Commodity microwave ovens output a continuous wave (CW) with a fixed power that often exceeds 1000 W. They cannot be directly used for PDMS curing since the silicon would absorb most of the energy and the resulting rapid heat generation would cause an intolerable temperature rise.

One solution to this bottleneck could be to use variable power output microwave generators, but their cost is too high to be universally accessible.

The solution proposed here utilizes pulse width modulation (PWM)—the method of supplying energy in pulses so as to precisely control the average power being supplied. In this mode, the microwave output is turned on only for a specific duration, called the ON period. In this period, the silicon absorbs almost all the microwave energy and converts it into heat. Thereafter, no irradiation is allowed for another fixed duration, called the OFF period. In this period, the heat generated in the silicon dissipates into the PDMS layer via conduction. Thus, the temperature is not allowed to build up to destructive levels, but only sufficiently high to rapidly cure the PDMS. To ensure an upper bound on the temperature, the ON time of the pulse needs to be bound as well. If it is too high, the thermal energy still accumulates rapidly, leading to exceedingly high temperatures that degrade the master device. Ideally, the control over the microwave output should be possible with a high time resolution. However the transients in the microwave power circuitry do not allow sub-second control, as observed experimentally. Hence, for the purpose of experimenting with different average power levels, a fixed ON period of 1 second has been utilized. The overall power has been varied by changing the OFF period of the microwave irradiation. The degree of this variation can be denoted by the term 'duty cycle'. It is a dimensionless number that can vary between 0.0 and 1.0 and is calculated as

$$\text{duty cycle} = \frac{\text{ON}_{\text{time}}}{(\text{ON}_{\text{time}} + \text{OFF}_{\text{time}})}. \quad (2)$$

Table 1 Dielectric constant component values

Material	Real part (ϵ')	Imaginary part (ϵ'')
Silicon (Siddharth 2014)	11.7	64.0
PDMS (Awang et al. 2016)	2.9	0.04
Glass (Young and Freedman 2008)	3.7	0.02

2 Simulations and experimental setup

The common laboratory protocol for PDMS-based device fabrication is to place the master into a glass Petri dish, on to which the degassed PDMS prepolymer is poured and then thermally cured. This setup was replicated in the simulation

and experimental work detailed in the following text. However, the minute structures present on the surface of the master are not considered here. This is since they are most often constructed out of the polymer Su-8 which has a thermal conductivity (0.2 W/m K) (Lorenz et al. 1998) closely matched to that of PDMS (0.15 W/m K) (Datasheet 2017) and thus would thermally behave as a part of the PDMS layer itself. Moreover, the volume of Su-8 structures is also negligible compared to that of other components since the depth is usually less than 50 μm and the area fill factor is also quite low. Thus, including the Su-8 structures would only increase the computational complexity without any useful gain in the results.

2.1 Simulation

Simulations were aimed at gauging the time frames necessary for the bulk PDMS to reach a temperature of at least 70 °C throughout its thickness. This is a temperature commonly employed in hot-air curing. Lower temperatures would prolong the curing process while much higher temperatures are known to degrade the mechanical properties of PDMS (Datasheet 2017). Thus, this temperature of 70 °C is a comfortably placed reference.

Simulations were performed using finite element modeling (FEM) software COMSOL. A geometric model corresponding to the actual physical setup was constructed. A multiphysics simulation was setup, comprising of electromagnetic and solid-state heat transfer simulation. While the former assesses the thermal energy generation, the latter uses this information to build temporal temperature profiles. The electromagnetic simulation was based on Maxwell's equations, culminating in the form presented as follows:

$$\nabla \times (\mu_r^{-1} \nabla \times E) = k_0^2 \left(\epsilon_r - \frac{i\sigma}{\omega\epsilon_0} \right) \vec{E}, \quad (3)$$

where μ_r is the magnetic permeability, $\vec{E}$ is electric field intensity in the mesh of interest, k_0 is the refractive index, σ is the electrical conductivity, ω is the frequency of the electromagnetic field, and ϵ_0 is the electric permittivity of vacuum.

Equation 1, along with Eq. 3, yields the thermal energy generated per unit time, 'Q'. The heat transfer simulation then uses the following equation to build the temperature profile across the structures as a function of time:

$$\rho C_p \frac{\delta T}{\delta t} = \nabla \cdot (k \nabla T) + Q, \quad (4)$$

where ρ is the density, C_p is the heat capacity, T is the temperature, t is the time, k is the thermal conductivity, and Q is the thermal energy per unit time.

Boundary conditions and mesh structure Since the internal surface of the microwave oven are usually made of a highly conductive material, the resistive energy loss at the walls is insignificant. Correspondingly, the boundary condition set was $\vec{n} \times \vec{E} = 0$, implying the tangential electric field component to be zero. The heat transfer was applied only for the domains of silicon, PDMS and glass and thus the heat loss to ambient is neglected. This is a plausible assumption since the microwave oven is a closed cavity and thus heat loss is limited, and also this heat loss is insignificant to the heat flux between the three modelled domains. A mirror symmetry was used to reduce computation time, by vertically slicing the model shown in Fig. 4, midway. The corresponding boundary condition was thus $\vec{n} \times \vec{H} = 0$. For both the simulations, the mesh was set to the default tetrahedral mesh, with about 136,000 elements.

The geometric model had four entities: (a) the glass turntable plate (for even distribution of incident microwave energy), (b) the glass Petri dish placed on the turntable plate, (c) the silicon wafer kept in the Petri dish and (d) the poured PDMS on the silicon wafer (and contained by the Petri dish). A simulation schema containing these four entities was built in the FEM software. Figure 1 denotes a cross-sectional view of this simulation design.

The dimensions of these entities as well as other relevant parameters for both sets of simulations are stated in Table 2. The PDMS layer thickness was varied between 1 and 10 mm, though about 5 mm is enough for inlet and outlet tubings to be connected with sufficient mechanical support to avoid leakages. It is thus a commonly used PDMS thickness for microfluidic devices. The silicon wafer diameter was fixed at 4 in. and the entire silicon surface was assumed to be occupied by the PDMS layer. The other parameters such as turntable plate radius and microwave power output are chosen as per the actual values pertaining to the microwave

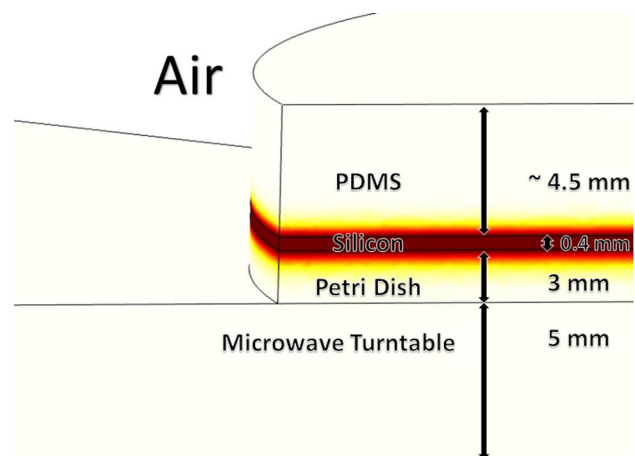


Fig. 1 Cross-sectional view of the simulation design, modelled after the actual physical setup

Table 2 Simulation parameters

Parameter	Description	Value
r_{si}	Silicon wafer radius	51 mm
t_{si}	Silicon wafer thickness	0.4 mm
r_{p}	PDMS radius	51 mm
t_{p}	PDMS thickness	4.5 mm
r_{p}	Glass turntable plate radius	122 mm
t_{p}	Glass turntable plate thickness	5 mm
p_{uW}	Unmodulated microwave power	700 W
f_{uW}	Microwave frequency	2450 MHz
T_0	Initial PDMS temperature	25 °C

oven (Bajaj 1701 MT; price approximately US \$60) used in the experiments.

The pulsed microwave power was simulated by multiplying the unmodulated microwave power output (p_{uW}) with a rectangular pulse function. The average power delivered was varied by changing the duty cycle of the pulse function. By carefully adjusting the duty cycle, it was ensured that (a) the system temperature does not increase beyond tolerable limits, and (b) the thermal energy gets sufficient time to dissipate through the PDMS, curing it in the process. To vary the duty cycle, the ON period was kept constant at 1 s and the OFF period length was varied.

2.2 Experimental setup

The internal circuit of the microwave oven (Bajaj 1701 MT; 700 W power) was modified to gain control over the irradiation. This aided the experimentation and helped achieve an optimum curing protocol. This modification can be easily adapted by manufacturers to build low-cost curing systems with the aid of the results presented in this work. Also, commercial microwave ovens may also be used directly without any modifications. This is so since they contain pre-programmed settings that allow pulsed irradiation with several different duty cycles. Some of them would match with the protocols reported here and could be used as such.

This control was implemented here by a low-cost, custom-built control circuit, which comprised of a commonly available micro-controller based digital processing board (Arduino Uno Rev. 3) and an electro-magnetic relay. Figure 2 shows a photograph of the setup and Fig. 3 shows the schematic of the control circuitry. The safety features such as thermal-cutoffs and fuses have not been shown for the sake of brevity. The soft lithography master was a generic design of a microfluidic device formed on a 4-in. silicon wafer. The structural material of the design was made of Su-8 (Dow Corning Su-8 2100).

For ascertaining the completion of the curing process, the samples were withdrawn from the microwave oven at

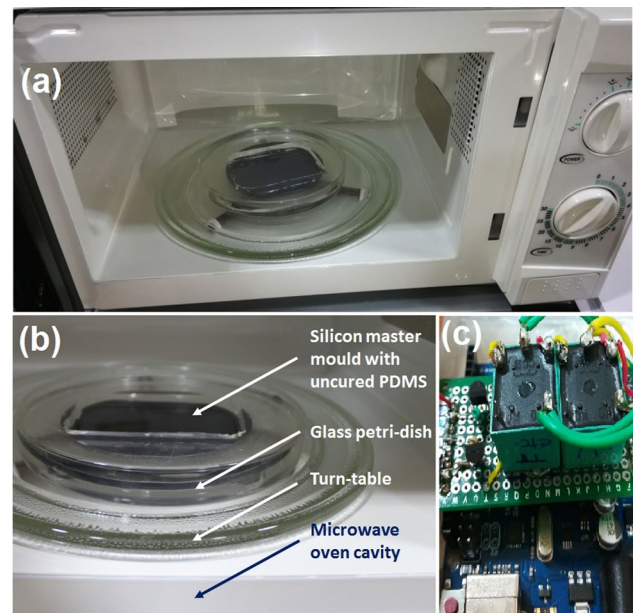


Fig. 2 Photograph of the setup. **a** The Petri dish placed inside the microwave oven, **b** close-up view of the Petri dish and **c** the power controlling components placed on top of the digital logic board

10-s intervals and the topmost surface was checked for phase change. The time at which the surface had hardened was considered as the optimal cure time. Once this time was established, a second repetition of the experiment was conducted. This time, samples were withdrawn only after the completion of the stipulated curing period as found in the first run. The results of the two experiments were in agreement, save for a small difference of less than 10 s. This difference was caused by the extra time added in first set of experiments due to the time spent in withdrawing, checking for curing and reinserting the sample. The cure times reported in this text are the results of the second set of experiments.

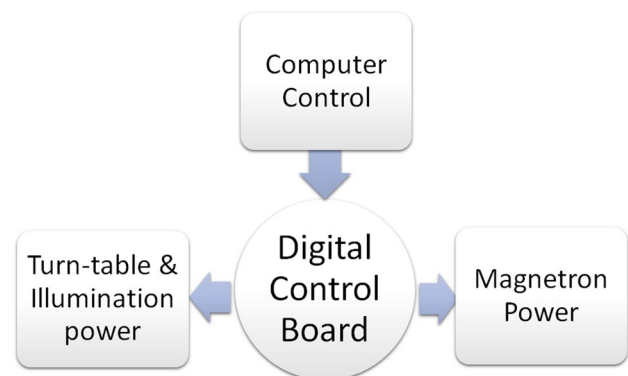


Fig. 3 Schematic of microwave power control circuitry

3 Results and discussion

3.1 The need for pulsed irradiation

For instance, consider a standard silicon wafer of thickness 0.4 mm containing a 5-mm (a commonly used thickness in microfluidic applications) layer of PDMS prepolymer. To cure this layer, its temperature has to be raised from the ambient 25 °C to 70 °C (the temperature generally targeted for a complete PDMS cure). For this process, an energy input of about 350 mJ/mm² will be needed (calculations depicted in Eq. 5). If the microwave oven is rated at 700 W (and 75% efficiency), it generates about 2.18×10^8 W/m³ or 46 mW/mm² of heat in the silicon wafer (as reported by FEM simulation). Thus, about 8 s (= 350/46) of continuous irradiation will be enough to impart this energy. However, as the simulation results in Fig. 4 show, this 8 s irradiation will result in a temperature of more than 250 °C at the Si–PDMS interface, which is well beyond the limit (200 °C) for preventing PDMS thermal degradation.

Thermal energy required for curing is given by ‘ q ’:

$$q = C \times (\rho V) \times \delta C, \quad (5)$$

where C is heat capacity of PDMS (1.46 J/g K), (ρV) is the mass of PDMS [where ρ is (965 g/l)], and δC is the temperature change.

For the standard values of C and ρ (Datasheet 2017) as noted, and δC of 45 °C (from ambient temperature to cure temperature of 70 °C), ‘ q ’ equals to 0.35 J/mm³.

At such high temperatures, a thin layer of PDMS (at the silicon–PDMS interface) polymerizes immediately and then starts breaking down into cyclic oligomers (Camino et al. 2002). These then destroy the master relief structures by irreversibly bonding to them. The use of silicon is

unavoidable because the microfabrication processes developed for silicon-based semiconductor devices have been well adapted to achieve reliable and high-resolution microfluidic master moulds.

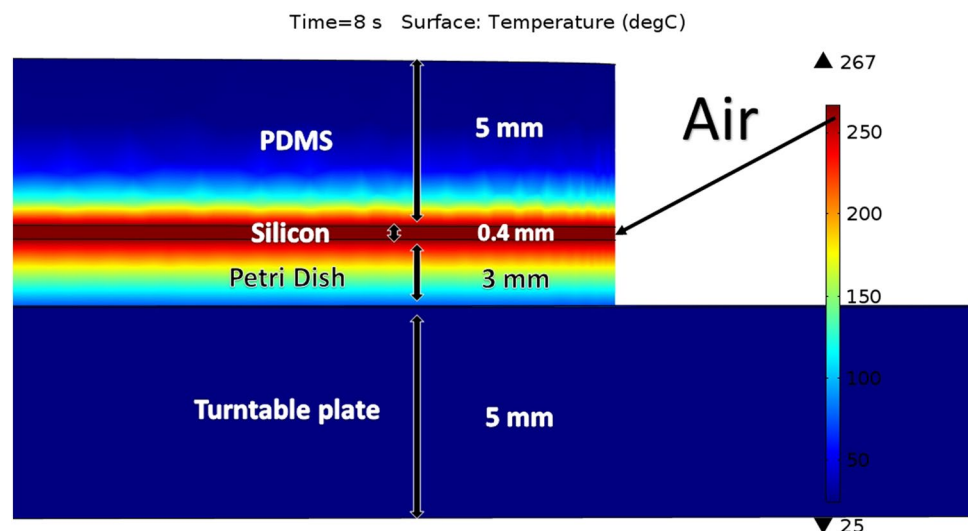
3.2 Simulation results for optimal duty cycles

The transient thermal profile throughout the PDMS layer was simulated under four different microwave power levels (700, 1000, 1250 and 1500 W) and these were pulsed at varying duty cycles (from 50 to 12.5%). The power levels correspond to the commonly available microwave powers in the commodity market. Further, the simulations were performed for a range of PDMS layer thicknesses –1 to 10 mm, with a step size of 1 mm. The simulations were aimed at ascertaining the smallest cure time without exceeding the PDMS thermal degradation temperature. This exhaustive dataset should be able to serve to the needs of majority of microfluidic labs.

Figures 5, 6, 7 and 8 show the minimum time (in seconds) needed for a PDMS layer to reach an average temperature of 70 °C at the top surface, for the four different microwave power levels. Since the lower surfaces would anyway be at a higher temperature, the whole of PDMS thickness would be cured in that time. However, high-duty cycles may cause the bottom layer of PDMS (PDMS–silicon interface) to rise beyond 200 °C, and thus the times reported in these images exclude such cases.

As can be noticed from Fig. 5, a duty cycle of 50% is expected to be useful for up to a 7-mm PDMS layer. The expected time taken for 5 mm is about 2 min, while a thin layer (1 mm) should get cured in a third of a minute. However, a 10-mm layer will need about 6 min, partly due to the lower duty cycle of 25%. These cure times change for a slightly higher microwave power of 1000 W (Fig. 6),

Fig. 4 Cross-sectional view of the simulation design, depicting the spatial temperature gradients after an 8-s continuous wave microwave (700 W) irradiation. Overshoots beyond 200 °C can be seen at the Si–PDMS interface



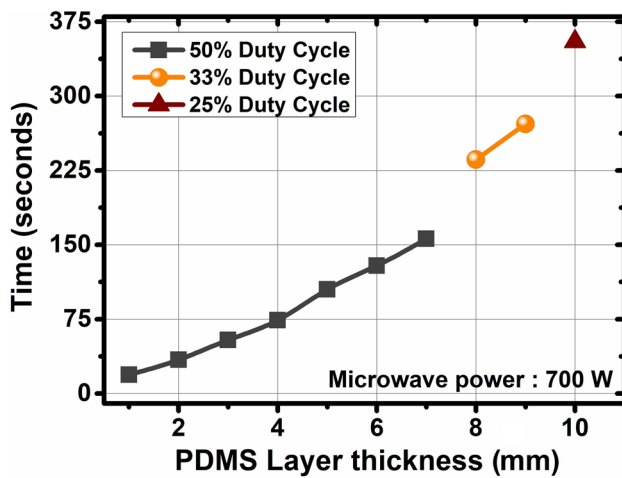


Fig. 5 Cure time as a function of PDMS thickness for a 700 W microwave

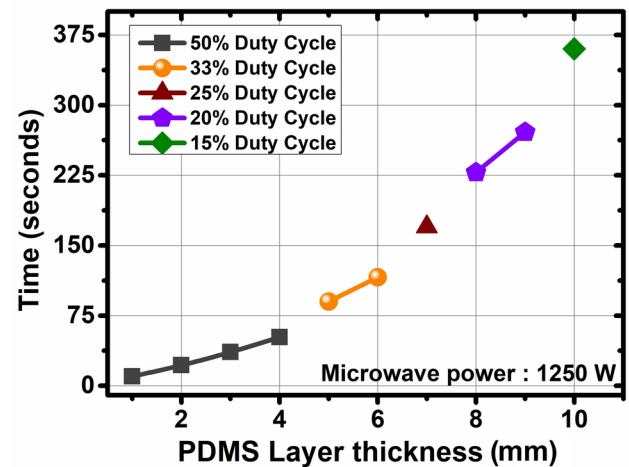


Fig. 7 Cure time as a function of PDMS thickness for a 1250 W microwave

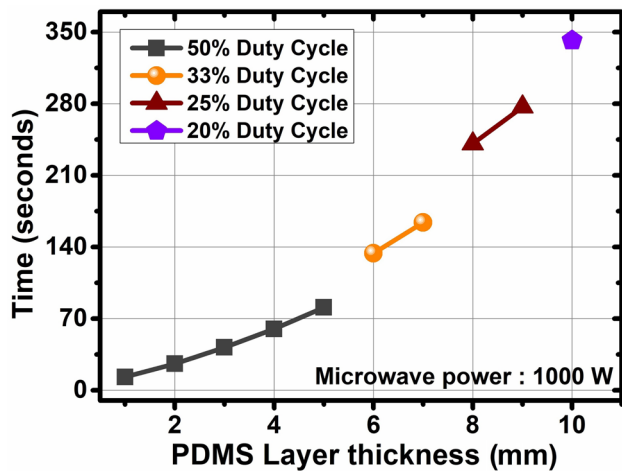


Fig. 6 Cure time as a function of PDMS thickness for a 1000 W microwave

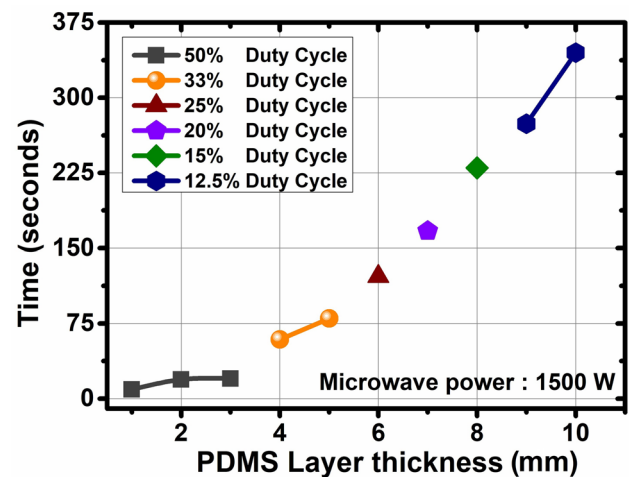


Fig. 8 Cure time as a function of PDMS thickness for a 1500 W microwave

with a 5-mm layer expected to be cured in about a 1.5 min and a 10-mm layer needing a few seconds less than for a 700 W oven. However, the cure time remains same or even slightly increases for thicknesses of 8 and 9 mm since these now need a lower duty cycle of 25% (instead of the 33% in case of a 700 W oven). Thus, increasing microwave power may not necessarily yield a faster cure and, in fact, can be detrimental. This trend is continued for higher powers, as can be seen in Figs. 7 and 8. The cure time for a 5-mm layer is again about 1.5 min for both, 1250 and 1500 W ovens. Similarly, for higher thicknesses of 9 and 10 mm, the cure time registers only slight changes. The reduction in cure time due to increasing power can be mainly noticed in smaller thicknesses of 1 and 2 mm, where a duty cycle of 50% is still applicable and thus the

benefits of higher microwave power are reflected directly in the cure time.

Figure 9 summarizes the effect of changing microwave power on cure times, for three different PDMS layer thicknesses. As can be noticed, cure time for a small PDMS thickness of 1 mm decreases from 19 to 9 s as the microwave power increases from 700 to 1500 W. However, for higher PDMS thicknesses, the simulated cure time fluctuates, with the minima reported at 100 W for both, 5- and 10-mm layers. This is again due to the need for lower duty cycles at high microwave power so as to provide sufficient time for the thermal energy to dissipate. To conclude, for a PDMS layer of 5 mm, a duty cycle of 50% may be used only if microwave power does not exceed 1000 W; otherwise it is necessary to reduce it to 33% or lower.

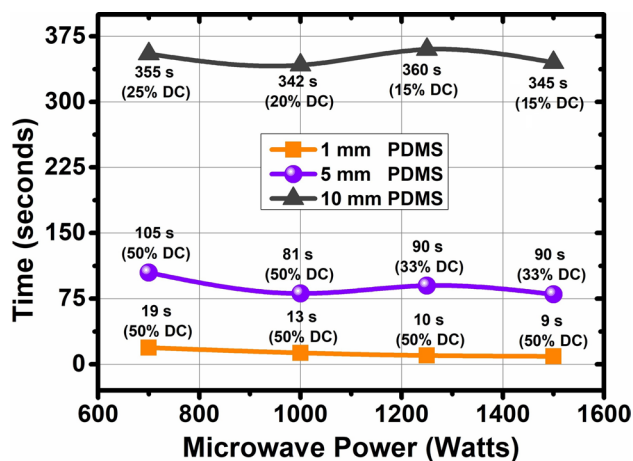


Fig. 9 Optimal cure time as a function of PDMS thickness for multiple microwave powers

3.3 Experimental results

Multiple samples of 5 mm thick layer of PDMS were cured at different duty cycles. The cure times are depicted in Fig. 10 and they are well in agreement with the results predicted by simulations. For all the experiments, Dow Corning Sylgard 184 with a base to curing agent ratio of 10:1 was used. After exposing to microwave radiation, the glass Petri dish containing the master and the PDMS was allowed to cool down for 5 minutes. For the purpose of comparative analysis, a sample of PDMS was also cured in a hot-air oven at 70 °C for 2 h. Thereafter the device was cut, punched and bonded via air plasma bonding to a glass slide. The microwave cured devices demonstrated no bonding issues.

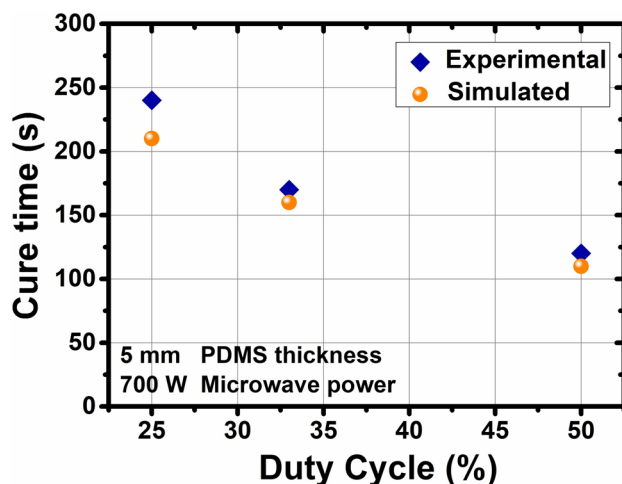


Fig. 10 Cure time as a function of duty cycle (700 W microwave power; 5 mm PDMS thickness)

4 Optical and mechanical characterization

The primary requirements from the PDMS-based microfluidic devices are to be mechanically stable for fluid flow and optically clear to allow opto-interrogation. To ascertain the effect of microwave curing on these two requirements, characterization experiments were performed. UV–Vis absorption spectral measurements were performed using a spectrophotometer (DeNovix) and tensile testing using a micro-UTM (Universal Testing Machine). The bulk PDMS samples ranged in thickness from 3.9 to 4.5 mm.

The optical data (presented in Fig. 11) revealed consistently low absorption at all optically relevant wavelengths (300–750 nm) amongst the differently cured samples. Only a slight variation in the intensity of absorption was caused due to the slight variation in the sample thicknesses that caused varying optical path lengths. Thus, the microwave cured PDMS is optically at par with hot-air-cured ones. The tensile tests were conducted after preparing samples as per the ASTM D412 Type C standard. The data, plotted in Fig. 12, reveal that the strength of microwave cured PDMS samples is close to that of the hot-air-cured one. The slight disparity in the results can be attributed to the higher temperatures experienced by PDMS as the duty cycle (and hence the average power delivered) increases. This is so as higher cure temperatures have been known to degrade the material strength of PDMS (Datasheet 2017). However, for all practical purposes, these variations are not important as the average stress and strain experienced in microfluidic devices under flow conditions are quite lower. For instance, even in a shallow channel (which is expected to be more prone to channel bulging) of 30 μm depth and 500 μm width, the highest deformation was reported to be 24 μm at a rather high flow rate of 800 $\mu\text{l}/\text{min}$ by Gervais et al. (2006). In

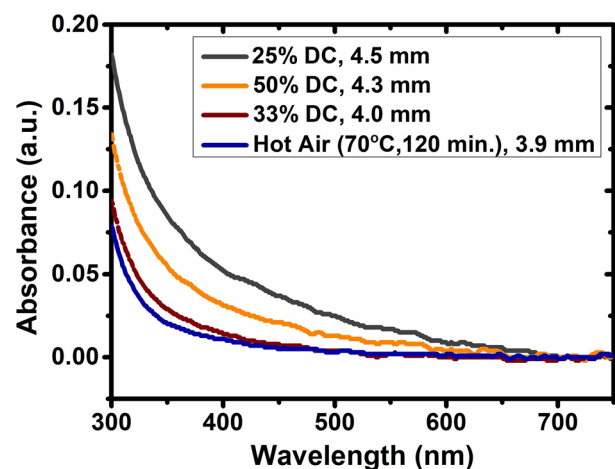


Fig. 11 UV–Vis absorbance of differently cured PDMS samples

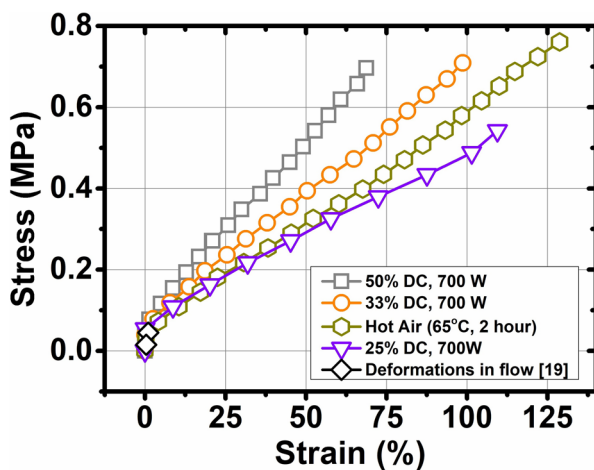


Fig. 12 Tensile testing results depicting similar characteristics of differently cured PDMS samples

yet another study, a channel of 37 μm depth and 500 μm width experienced a maximum strain of 1% at a stress of 0.045 MPa and a minimum strain of 0.4% at a stress of 0.015 MPa (plotted in Fig. 12), caused by a pressure-driven flow at the rate of 200 $\mu\text{l}/\text{min}$ and 50 $\mu\text{l}/\text{min}$, respectively. Thus, the microwave-cured PDMS is expected to be very well capable of enduring a wide range of flow conditions.

5 Conclusions

Rapid curing of PDMS would be an important step towards shortening the microfluidic device prototyping times. To this end, a specifically designed microwave-based curing protocol is presented here that allows a 2-min PDMS cure for the most commonly used microfluidic device thickness of about 5 mm. This represents more than an order of magnitude of reduction compared to conventional hot-air curing, which requires 120 min or more. A comprehensive simulation result set is presented which covers the commonly available microwave oven powers, expected PDMS thicknesses for microfluidic devices and the duty cycles required for the curing process. These results can serve as a guideline for choosing the duty cycle and irradiation time for the combination of microwave power (up to 1500 W) and device thickness (up to 10 mm), as required. It is highlighted that while larger duty cycles (50% or so) deliver energy at a higher rate, they may not always lead to shorter cure time, especially for higher device thicknesses (more than 5 mm). Thus, it is optimal to consult the simulation data presented here. Further, the mechanical and optical properties of PDMS are not affected by this method, as verified by the UV–Vis and tensile testing results.

The protocol can be readily implemented using low-cost, easily available commodity microwave ovens. Manufacturers can swiftly implement the electronic control with minimal changes to the circuitry and thus produce low-cost PDMS curing systems. Also, since commodity microwave ovens come pre-programmed with settings for pulsed irradiation, some of them could be used directly by matching their settings with the protocol presented here. Thus, this paper, for the first time, presents a theoretical basis, detailed simulations and experimental results for microwave-based rapid PDMS curing pertaining to replica moulding. Experiments reveal a close match between the simulated and experimental cure times for the most commonly used device thickness of 5 mm with a microwave irradiation power of 700 W. In addition, the optical and mechanical characterizations indicate suitability of this method for a wide range of microfluidic device requirements. Thus, the work presented here results in a time-efficient and reproducible PDMS curing protocol which can be a viable alternative to the traditional hot-air curing method. This method can be readily adopted by researchers and industry alike.

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Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

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